Reporte de Análisis Modal Espectral 2D

1. Propiedades del Pórtico

Se analizó un pórtico 2D con un módulo de elasticidad $E = 1.0$ T/m². Las columnas tienen un área $A = 1000000.0000$ m² y una inercia $I = 36826.875000$ m⁴. La viga presenta un área $A = 1000000000.0000$ m² y una inercia $I = 1000000000.000000$ m⁴.

2. Matrices Globales Condensadas

Matriz de Rigidez Lateral (K):

	0	1	2	3	4	5	6	7
0	200032520.25	−200000213.24	−16457.56	−90.06	28.17	26.67	379.82	379.82
1	−200000213.24	200032520.25	−90.06	−16457.56	26.67	28.17	379.82	379.82
2	−16457.56	−90.06	200032492.07	−200000239.92	−16483.81	−116.31	421.12	419.62
3	−90.06	−16457.56	−200000239.92	200032492.07	−116.31	−16483.81	419.62	421.12
4	28.17	26.67	−16483.81	−116.31	200032464.86	−200000267.12	−16022.34	343.66
5	26.67	28.17	−116.31	−16483.81	−200000267.12	200032464.86	343.66	−16022.34
6	379.82	379.82	421.12	419.62	−16022.34	343.66	200015038.04	−200001326.44
7	379.82	379.82	419.62	421.12	343.66	−16022.34	−200001326.44	200015038.04

Matriz de Masas (M):

	0	1	2	3	4	5	6	7
0	50.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	50.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	50.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	50.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	50.0	0.0	0.0	0.0
5	0.0	0.0	0.0	0.0	0.0	50.0	0.0	0.0
6	0.0	0.0	0.0	0.0	0.0	0.0	50.0	0.0
7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	50.0

3. Propiedades Dinámicas

	w (rad/s)	V (w^2)	T (s)
1	5.6121	31.4961	1.1196
2	16.7257	279.7507	0.3757
3	27.3697	749.1027	0.2296
4	33.8972	1149.0203	0.1854
5	2828.4341	8000039.4723	0.0022
6	2828.485	8000327.2797	0.0022
7	2828.5629	8000768.2965	0.0022
8	2828.6315	8001156.1906	0.0022

3.1 Modos de Vibración Expandidos (φ)

	0	1	2	3	4	5	6	7
0	−0.0194	−0.0568	−0.0672	0.0434	0.0228	−0.0577	0.0657	−0.0429
1	−0.0026	0.0029	0.0002	−0.0019	0.0	0.0	−0.0	0.0
2	0.001	−0.0012	−0.0001	0.0008	−0.0	0.0	0.0	−0.0
3	−0.0194	−0.0568	−0.0672	0.0434	−0.0228	0.0577	−0.0657	0.0429
4	0.0026	−0.0029	−0.0002	0.0019	−0.0	−0.0	0.0	−0.0
5	0.001	−0.0012	−0.0001	0.0008	0.0	−0.0	−0.0	0.0
6	−0.0393	−0.0603	0.0193	−0.0667	0.0429	−0.0577	−0.0228	0.0657
7	−0.0041	0.0078	−0.0002	−0.002	0.0	0.0	0.0	−0.0
8	0.0017	−0.0031	0.0001	0.0008	−0.0	−0.0	0.0	0.0
9	−0.0393	−0.0603	0.0193	−0.0667	−0.0429	0.0577	0.0228	−0.0657
10	0.0041	−0.0078	0.0002	0.002	−0.0	−0.0	−0.0	0.0
11	0.0017	−0.0031	0.0001	0.0008	0.0	0.0	−0.0	−0.0
12	−0.0569	−0.007	0.059	0.0569	0.0577	−0.0	−0.0577	−0.0577
13	−0.0049	0.0118	−0.0044	−0.0026	0.0	0.0	0.0	0.0
14	0.002	−0.0047	0.0018	0.001	−0.0	−0.0	−0.0	0.0
15	−0.0569	−0.007	0.059	0.0569	−0.0577	0.0	0.0577	0.0577
16	0.0049	−0.0118	0.0044	0.0026	−0.0	−0.0	−0.0	−0.0
17	0.002	−0.0047	0.0018	0.001	0.0	0.0	0.0	−0.0
18	−0.0696	0.0556	−0.0404	−0.0209	0.0657	0.0577	0.0429	0.0228
19	−0.0051	0.0132	−0.0071	−0.0047	0.0	−0.0	−0.0	0.0
20	0.002	−0.0053	0.0028	0.0019	−0.0	−0.0	−0.0	−0.0
21	−0.0696	0.0556	−0.0404	−0.0209	−0.0657	−0.0577	−0.0429	−0.0228
22	0.0051	−0.0132	0.0071	0.0047	−0.0	−0.0	−0.0	−0.0
23	0.002	−0.0053	0.0028	0.0019	0.0	0.0	0.0	0.0

4. Participación Modal

	Lw	Masa Efectiva (%)	Acumulada (%)
1	−18.5159	85.71	85.71
2	−6.8478	11.72	97.43
3	−2.943	2.17	99.6
4	1.2681	0.4	100.0
5	0.0	0.0	100.0
6	0.0	0.0	100.0
7	0.0	0.0	100.0
8	−0.0	0.0	100.0

5. Respuestas Máximas y Superposición

5.1 Desplazamientos Modales Máximos ($q_{\{\max\}}$)

	0	1	2	3	4	5	6	7
0	0.111573	0.013628	0.002591	0.000469	0.0	−0.0	0.0	0.0
1	0.014766	−0.0007	−6e-06	−2.1e-05	0.0	0.0	−0.0	−0.0
2	−0.00591	0.00028	3e-06	8e-06	−0.0	0.0	0.0	0.0
3	0.111573	0.013628	0.002591	0.000469	−0.0	0.0	−0.0	−0.0
4	−0.014766	0.0007	6e-06	2.1e-05	−0.0	−0.0	0.0	0.0
5	−0.00591	0.00028	3e-06	8e-06	0.0	−0.0	−0.0	−0.0
6	0.226713	0.014481	−0.000742	−0.000721	0.0	−0.0	−0.0	−0.0
7	0.02383	−0.001882	9e-06	−2.1e-05	0.0	0.0	0.0	0.0
8	−0.009535	0.000753	−3e-06	9e-06	−0.0	−0.0	0.0	−0.0
9	0.226713	0.014481	−0.000742	−0.000721	−0.0	0.0	0.0	0.0
10	−0.02383	0.001882	−9e-06	2.1e-05	−0.0	−0.0	−0.0	−0.0
11	−0.009535	0.000753	−3e-06	9e-06	0.0	0.0	−0.0	0.0
12	0.328066	0.001672	−0.002271	0.000616	0.0	−0.0	−0.0	0.0
13	0.028146	−0.002842	0.000169	−2.8e-05	0.0	0.0	0.0	−0.0
14	−0.011261	0.001137	−6.8e-05	1.1e-05	−0.0	−0.0	−0.0	−0.0
15	0.328066	0.001672	−0.002271	0.000616	−0.0	0.0	0.0	−0.0
16	−0.028146	0.002842	−0.000169	2.8e-05	−0.0	−0.0	−0.0	0.0
17	−0.011261	0.001137	−6.8e-05	1.1e-05	0.0	0.0	0.0	0.0
18	0.401117	−0.013343	0.001556	−0.000226	0.0	0.0	0.0	−0.0
19	0.029276	−0.003176	0.000274	−5.1e-05	0.0	−0.0	−0.0	−0.0
20	−0.011711	0.001271	−0.00011	2e-05	−0.0	−0.0	−0.0	0.0
21	0.401117	−0.013343	0.001556	−0.000226	−0.0	−0.0	−0.0	0.0
22	−0.029276	0.003176	−0.000274	5.1e-05	−0.0	−0.0	−0.0	0.0
23	−0.011711	0.001271	−0.00011	2e-05	0.0	0.0	0.0	−0.0

5.2 Fuerzas Estáticas Equivalentes ($Q_{\{\max\}}$)

	0	1	2	3	4	5	6	7
0	175.7049	190.6282	97.0404	26.9614	0.0	−0.0	0.0	0.0
1	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
2	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
3	175.7049	190.6282	97.0404	26.9614	−0.0	0.0	−0.0	−0.0
4	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
5	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
6	357.029	202.5498	−27.7961	−41.4478	0.0	−0.0	−0.0	−0.0
7	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
8	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
9	357.029	202.5498	−27.7961	−41.4478	−0.0	0.0	0.0	0.0

	0	1	2	3	4	5	6	7
10	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
11	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
12	516.6397	23.382	−85.069	35.3823	0.0	−0.0	−0.0	0.0
13	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
14	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
15	516.6397	23.382	−85.069	35.3823	−0.0	0.0	0.0	−0.0
16	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
17	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
18	631.6817	−186.6297	58.2945	−13.0115	0.0	0.0	0.0	−0.0
19	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
20	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
21	631.6817	−186.6297	58.2945	−13.0115	−0.0	−0.0	−0.0	0.0
22	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0
23	−0.0	−0.0	−0.0	0.0	0.0	0.0	0.0	−0.0

5.3 Fuerzas Combinadas (Regla SRSS)

	Fuerzas SRSS
0	196.7115
1	0.0
2	0.0
3	183.8571
4	0.0
5	0.0
6	356.1332
7	0.0
8	0.0
9	337.0283
10	0.0
11	0.0
12	506.4225
13	0.0
14	0.0
15	491.6065
16	0.0
17	0.0
18	661.3773
19	0.0
20	0.0
21	661.3773

	Fuerzas SRSS
22	0.0
23	0.0

6. Visualización de la Deformada

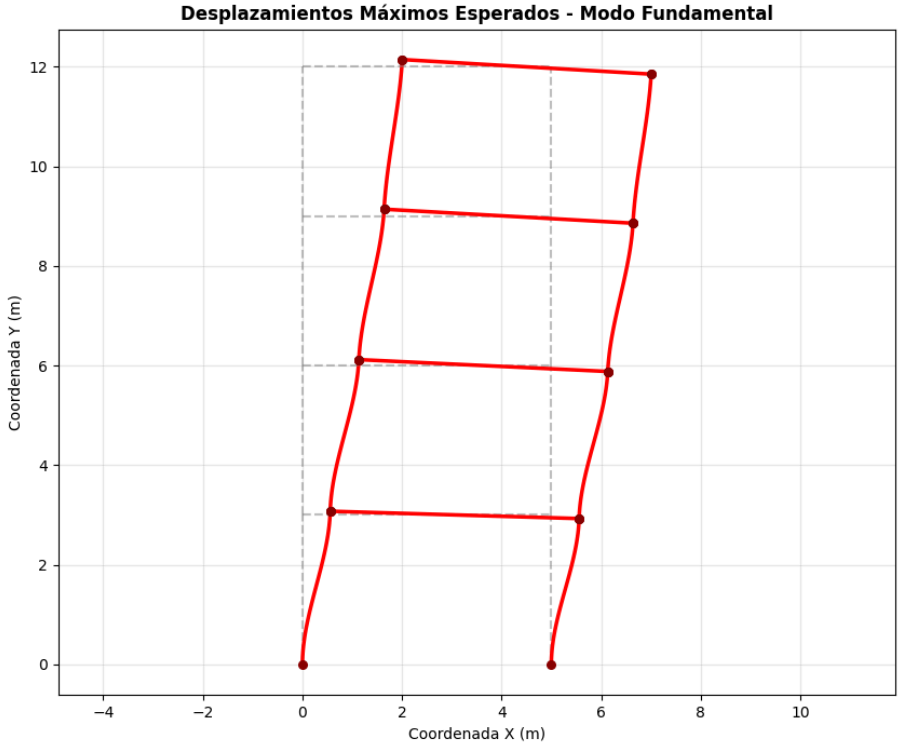

Figure 1: Desplazamientos máximos esperados del Modo Fundamental.

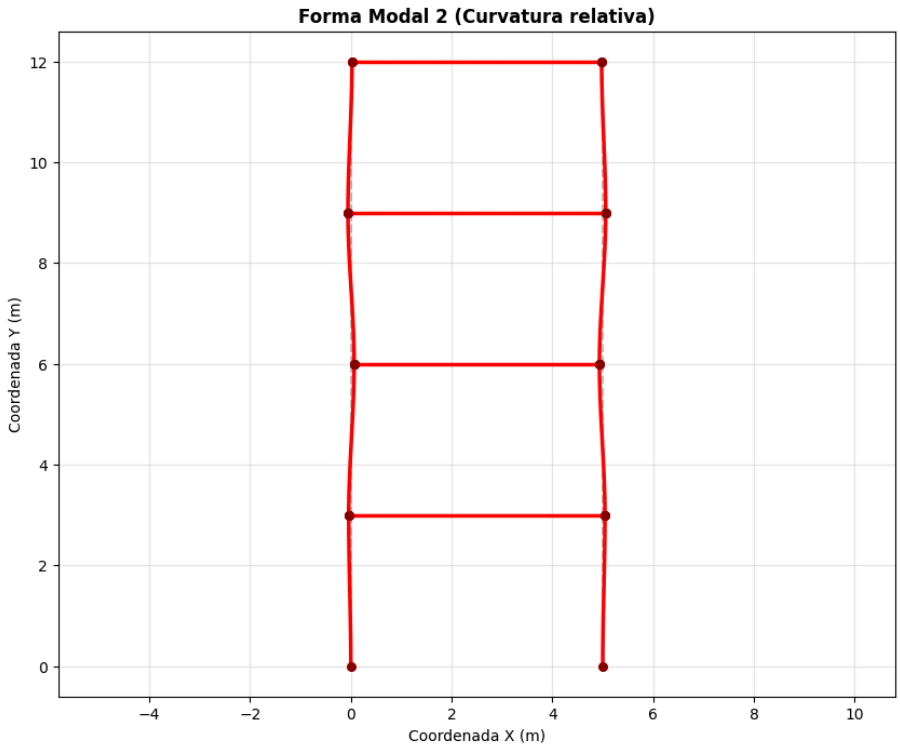

Figure 2: Forma Modal 2 (Curvatura relativa).